

(1)



$$J_{\text{pulley}} = \int_0^R \frac{2m}{\pi R^2} \cdot 2\pi r dr r^2$$

$$= \frac{2m}{\pi R^2} \cdot 2\pi \cdot \frac{1}{4} R^2$$

$$= mR^2$$

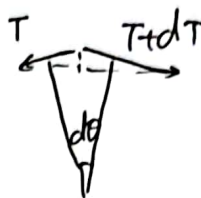
$$(J_{\text{pulley}} + J_1 + J_2)\beta = (2m - m)gR$$

$$\beta (mR^2 + 2mR^2 + mR^2) = mgR$$

$$\beta = \frac{g}{4R}$$

$$a = \beta R = \frac{1}{4}g$$

(2)



$$dT = f = \mu_0 N$$

$$N = T \frac{d\theta}{2} + (T + dT) \frac{d\theta}{2} = 2T \frac{d\theta}{2} = T d\theta$$

$$\mu_0 T d\theta = dT$$

$$\int_0^\pi \mu_0 d\theta = \int_{T_1}^{T_2} \frac{dT}{T}$$

$$\mu_0 \pi = \ln \frac{T_2}{T_1}$$

$$\begin{cases} 2mg - T_2 = 2ma \\ T_1 - mg = ma \\ a = \frac{1}{4}g \end{cases} \Rightarrow$$

$$T_2 = \frac{3}{2}mg$$

$$T_1 = \frac{5}{4}mg$$

$$\mu_0 \pi = \ln \frac{6}{5}$$

$$\mu_0 = \frac{1}{\pi} \ln \frac{6}{5}$$

(3)

$$dT = f = \frac{1}{2} \mu_0 N = \frac{1}{2} \mu_0 T d\theta$$

$$\frac{1}{2} \mu_0 \pi = \ln \frac{T_2'}{T_1'} = \frac{1}{2} \ln \frac{T_2'}{T_1'} = \frac{1}{2} \ln \frac{6}{5}$$

$$\frac{T_2'}{T_1'} = \sqrt{\frac{6}{5}}$$

$$\begin{cases} 2mg - T_2' = 2ma \\ T_1' - mg = ma \\ \frac{T_2'}{T_1'} = \sqrt{\frac{6}{5}} \end{cases}$$

$$\Rightarrow a = \frac{2\sqrt{5} - \sqrt{6}}{2\sqrt{5} + \sqrt{6}} g$$



二.

0.  $\frac{1}{4}\lambda$  P

11)  $y = A \cos[2\pi(\nu t + \frac{x}{\lambda}) + \pi]$   
 $= -A \cos[2\pi(\frac{x}{\lambda} + \nu t)]$

P 在  $x = \frac{1}{4}\lambda$  处

$y(P) = -A \cos[2\pi\nu t + \frac{1}{2}\pi]$   
 $= -A \sin(2\pi\nu t)$

(12) 缺失

(13) 驻波

$y_+ + y_- = 0 \cdot A \cos[2\pi(\nu t - \frac{x}{\lambda})]$   
 $- A \cos[2\pi(\nu t + \frac{x}{\lambda})]$   
 $= 2A \sin(2\pi\nu t) \sin(2\pi\frac{x}{\lambda})$

$x_0 = \frac{\pi}{8}$

$y(D) = \sqrt{3} A \sin(2\pi\nu t)$

三.

$\int \frac{dN}{N} = \int_0^{v_0} A v^2 dv$

(1)  $1 = A \cdot \frac{1}{3} v^3 \Big|_0^{v_0}$

$1 = A \cdot \frac{1}{3} v_0^3$

$A = \frac{3}{v_0^3}$

(2)  $A = \frac{3}{v_0^3} \bar{v} \quad (1)$

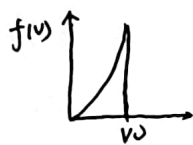
(3)  $v_p = v_0$

$\bar{v} = \int \frac{v dN}{N} = \int_0^{v_0} v f(v) dv = \frac{3}{4} v_0$

$\bar{v}^2 = \int \frac{v^2 dN}{N} = \int_0^{v_0} v^2 f(v) dv = \frac{3}{5} v_0^2$

$\sqrt{\bar{v}^2} = \sqrt{\frac{3}{5}} v_0$

$f(v) dv = \frac{dN}{N} \quad f(v) = \frac{3}{v_0^3} v^2$



(4)

$\bar{v}^2 = \int_0^{v_0} v^2 f(v) dv$

$= \int_0^{v_0} \frac{v^2 dN}{\frac{1}{3} N (0 - \frac{1}{3} v_0)}$

$= \frac{\int_0^{v_0} v^2 dN}{N' / N}$

$= \frac{\int_0^{v_0} \frac{v^2 dN}{N}}{\int_0^{v_0} \frac{dN}{N}} = \frac{\frac{3}{5} \frac{v_0^2}{25}}{\frac{1}{3}}$

$= \frac{3 v_0^2}{5 \cdot 25 \cdot 2}$

$\sqrt{\bar{v}^2} = \frac{1}{2} \sqrt{\frac{3}{5}} v_0$



13

$$\begin{aligned}
 (1) \quad dI(r) &= \frac{e 2\pi r dr l}{2\pi/w} \\
 &= e l w r dr \\
 I(r, t) &= \int_0^r dI(r) \\
 &= \frac{1}{2} e l \beta t r^2 \\
 I(a, t) &= \frac{1}{2} e l a^2 \beta t
 \end{aligned}$$



$$\begin{aligned}
 (2) \quad \oint B dl &= \mu_0 \sum I_{\text{内}} \\
 [B(a) - B(0)]l &= \mu_0 \cdot \int_0^r dI(r) = \mu_0 \cdot \frac{1}{2} e l \beta t r^2 \\
 \text{其中 } B(a) &\text{可看作无数层线圈外表面的 } (L \gg a) \\
 B(a) &= 0 \\
 B(0) - B(a) &= B(0) = \frac{\mu_0}{2} e l a^2 \beta t \\
 B(r) &= \frac{\mu_0}{2} e l \beta t (a^2 - r^2)
 \end{aligned}$$

(3)

$E_n$  法向由  $\rho$  产生

$$\begin{aligned}
 \oint E_n ds &= \frac{\sum q_{\text{内}}}{\epsilon_0} = \rho \pi r^2 l \\
 E_n 2\pi r l &= \frac{1}{2} e \pi r^2 l \\
 E_n &= \begin{cases} \frac{e r l}{2\epsilon_0} & (r < a) \\ \frac{e a^2 l}{2\epsilon_0 l} & (r > a) \end{cases} \Rightarrow \\
 E_n &= \frac{e a^2 l}{2\epsilon_0 l} \quad (r > a)
 \end{aligned}$$

$E_t$  切向由  $\frac{\partial B}{\partial t}$  产生

$$\begin{aligned}
 E \cdot 2\pi r &= - \oint \frac{\partial B}{\partial t} ds \\
 &= - \frac{\mu_0 e l \beta}{2} \int_0^r (a^2 - r^2) 2\pi r dr \\
 &= - \mu_0 \pi e l \beta \left( \frac{a^2 r^2}{2} - \frac{r^4}{4} \right) \\
 E &= \frac{-\mu_0 e l \beta}{2} \left( \frac{a^2 r^2}{2} - \frac{r^4}{4} \right) \\
 &= \frac{-\mu_0 e l \beta}{8} (2a^2 r^2 - r^4) \\
 &= - \frac{\mu_0 e l \beta}{8} (2a^2 r^2 - r^4)
 \end{aligned}$$



五.  
1)



$$(a+b)\sin\theta = k\lambda$$

$$2a = a+b \pm d = 500 \text{ nm} \quad \lambda = 590 \text{ nm}$$

$$\sin\theta = \frac{k\lambda}{d} \leq 1$$

$$k < \frac{d}{\lambda} = 3.38$$

$$k = 0, \pm 1, \pm 2, \pm 3$$

~~asinθ = kλ~~

单缝衍射因子

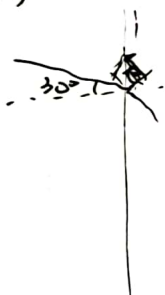
$$\left(\frac{\sin\alpha}{\alpha}\right)^2$$

$$\alpha = \frac{\pi}{\lambda} a \sin\theta = k'\pi \text{ 即 } a \sin\theta = k'\lambda \text{ 抵消}$$

或  $k=12$  时无意义

$$k = 0, \pm 1, \pm 3$$

12)



$$d(\sin\theta + \sin 30^\circ) = k\lambda$$

$$d(\sin\theta + \frac{1}{2}) = k\lambda$$

$$\sin\theta + \frac{1}{2} = \frac{k\lambda}{d}$$

$$\alpha \in (0, \pi) \\ \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$$

$$\frac{k\lambda}{d} \in [-\frac{1}{2}, \frac{3}{2}]$$

$$k \in (-\frac{d}{2\lambda}, \frac{3d}{2\lambda})$$

$$k = -1, 0, 1, 2, 3, 4, 5$$

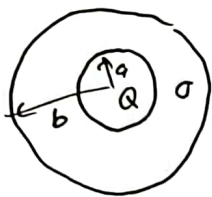
其中  $k=2, 4$  时  $a \sin\theta$  单缝衍射因子为 0

$$k = -1, 0, 1, 3, 5 \text{ 为有效}$$



★  
六

11)



a-b 间

$$E \cdot 4\pi r^2 = \frac{Q(t)}{\epsilon_0}$$

$$E = \frac{Q(t)}{4\pi r^2 \epsilon_0}$$

$$j = \sigma E$$

$$I = \frac{dQ}{dt} = jS$$

$$-\frac{dQ}{dt} = \frac{\sigma Q}{4\pi r^2 \epsilon_0} \cdot 4\pi r^2$$

$$\frac{dQ}{Q} = -\frac{\sigma}{\epsilon_0} dt$$

$$Q = Q_0 e^{-\frac{\sigma}{\epsilon_0} t} \quad Q_0 = Q$$

$$(2) \quad R = \int_a^b \frac{\rho dr}{4\pi r^2} = \frac{\rho}{4\pi} \frac{b-a}{ab}$$

$$dQ_{\text{壳}} = \int_0^t I R dt = \frac{\sigma}{\epsilon_0} Q(t) \cdot \frac{\rho}{4\pi} \frac{b-a}{ab} dt$$

$$Q_{\text{壳}} = \int_0^t \frac{\sigma Q_0}{\epsilon_0} e^{-\frac{\sigma}{\epsilon_0} t} \cdot \frac{\rho}{4\pi} \frac{b-a}{ab} dt$$

$$= \frac{\sigma Q_0^2 \rho}{8\pi \epsilon_0 ab} (1 - e^{-\frac{\sigma}{\epsilon_0} t}) e^{-\frac{1}{\sigma} t}$$

$$W = \int_{Q_0}^0 \frac{Q dQ}{2C} = \frac{1}{2C} (Q_0^2 - Q^2)$$

$$= \frac{1}{2} \frac{1}{4\pi \epsilon_0} \frac{b-a}{ab} Q_0^2 (e^{-\frac{\sigma}{\epsilon_0} t} - 1)$$

$$= \frac{Q_0^2}{8\pi \epsilon_0} \frac{b-a}{ab} (e^{-\frac{\sigma}{\epsilon_0} t} - 1)$$

t.

11)

$$E_1 4\pi r^2 = \frac{Q_1}{\epsilon_0}$$

$$E_1 = \frac{Q_1}{4\pi r^2 \epsilon_0}$$

$$E_2 = \frac{Q_2}{4\pi r^2 \epsilon_0 \epsilon_r}$$

电势相等  $\oint E dl = 0$

$$E_1 - E_2 = 0$$

$$E_1 = E_2 \quad \text{即}$$

$$Q_1 = \frac{Q_2}{\epsilon_r} \quad \text{--- ②}$$

$$\begin{cases} Q_1 + Q_2 = Q \\ Q_1 = \frac{Q_2}{\epsilon_r} \end{cases} \Rightarrow$$

$$\begin{cases} Q_1 = \frac{1}{\epsilon_r + 1} Q \\ Q_2 = \frac{\epsilon_r}{\epsilon_r + 1} Q \end{cases}$$

(2)

$$\Delta U = \int E_1 dr$$

$$= \frac{Q_1}{4\pi \epsilon_0} \int_{R_1}^{R_2} \frac{1}{r^2} dr$$

$$U_1 - U_2 = \frac{Q_1}{4\pi \epsilon_0} \frac{R_2 - R_1}{R_1 R_2} = \frac{Q}{4\pi \epsilon_0 (\epsilon_r + 1)} \frac{R_2 - R_1}{R_1 R_2}$$

(3)

$$U_1 = U_2 + \Delta U$$

$$= \frac{Q}{4\pi \epsilon_0 R_2} + \frac{Q}{4\pi \epsilon_0 (\epsilon_r + 1)} \frac{R_2 - R_1}{R_1 R_2}$$

